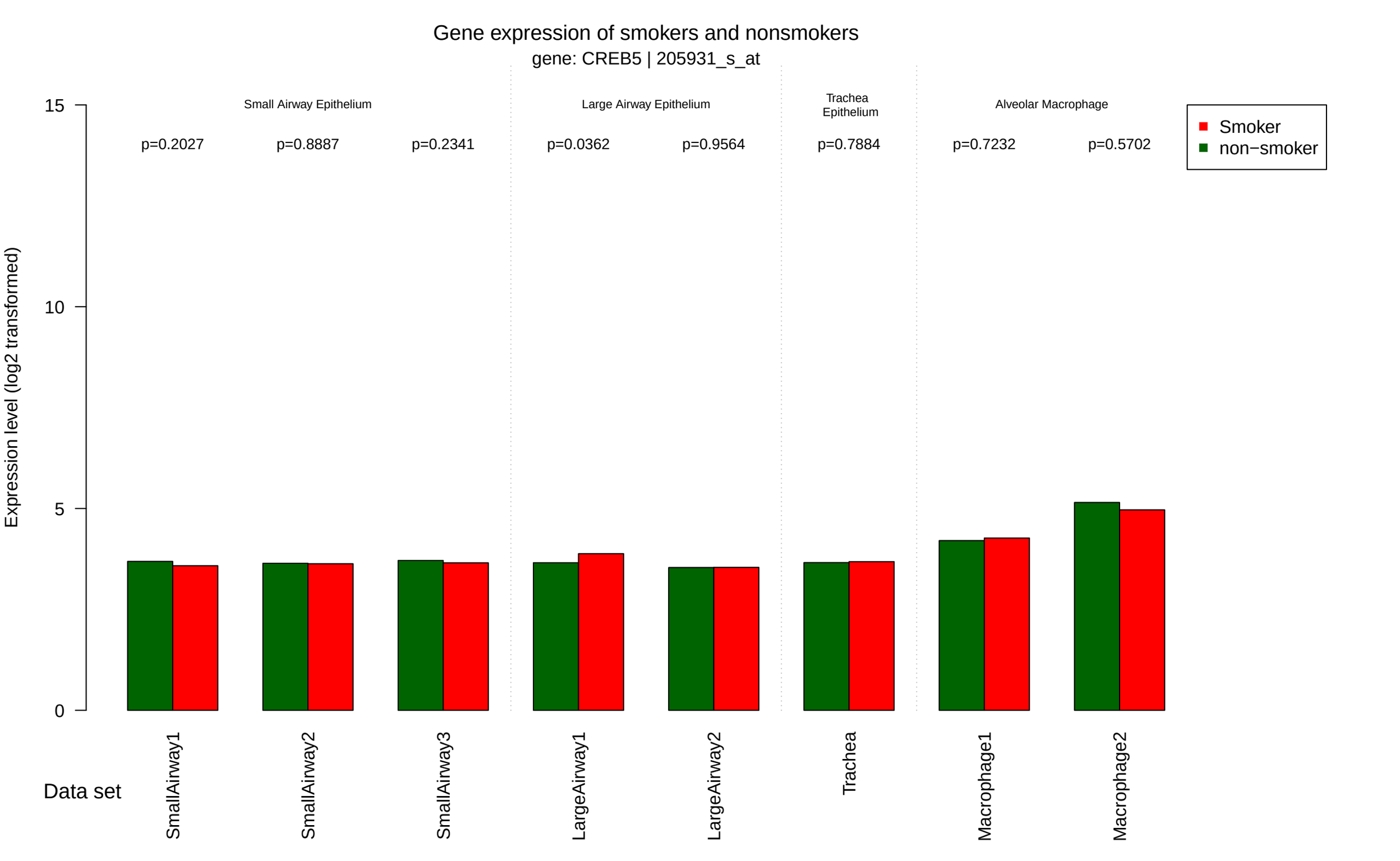
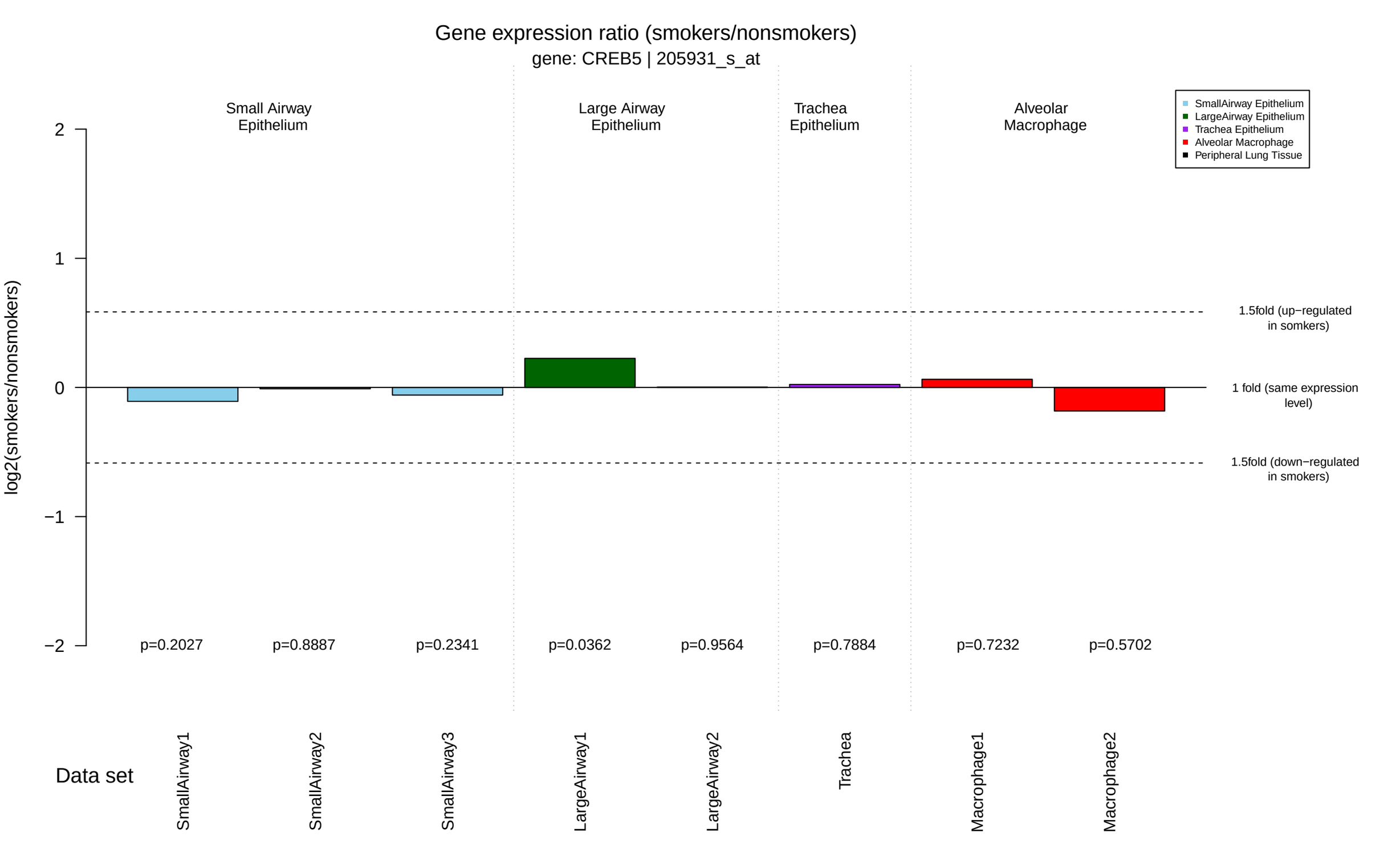
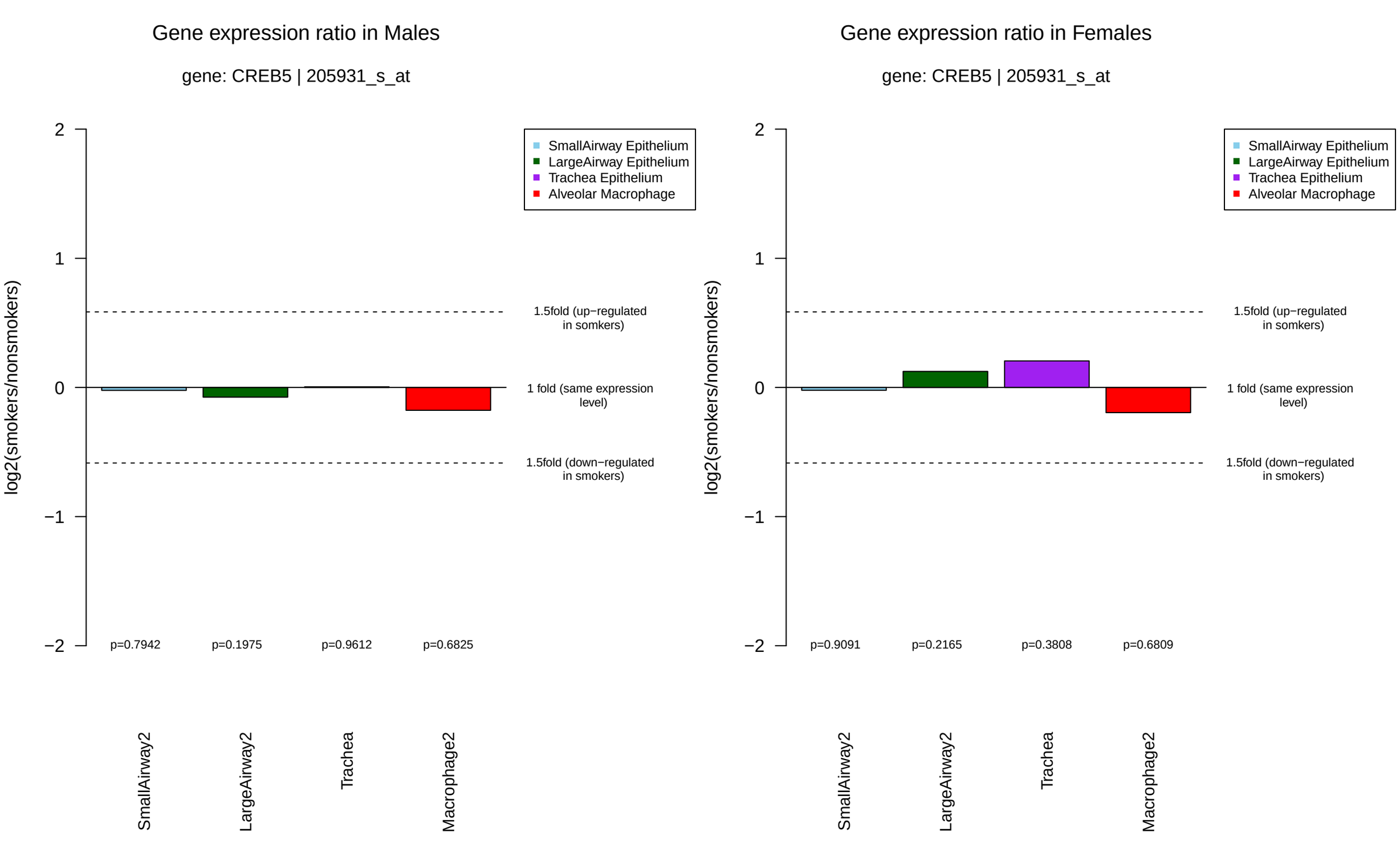
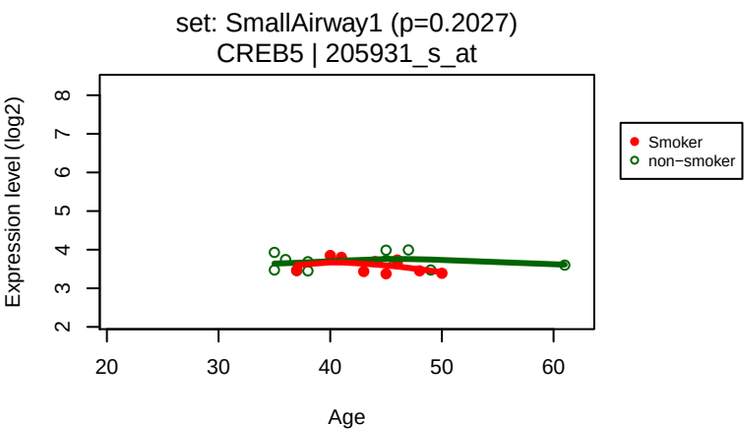
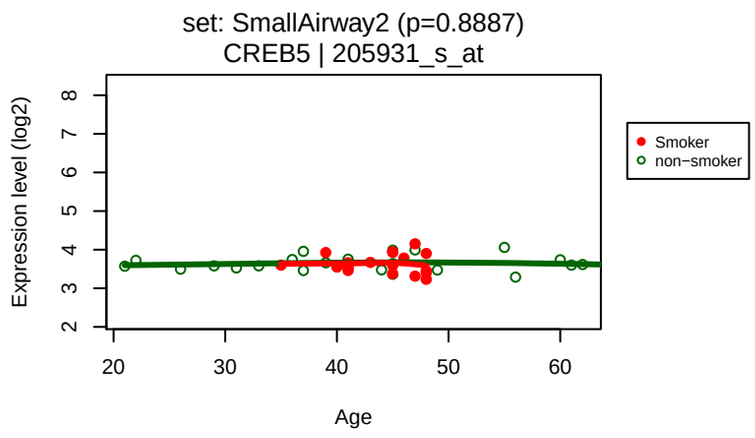
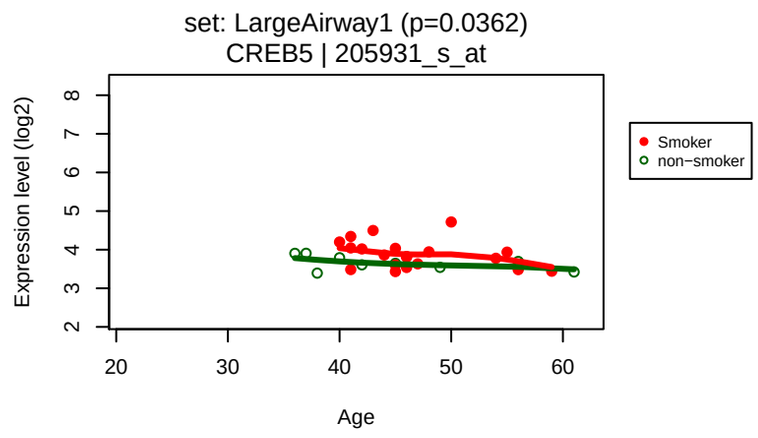

Gene expression of smokers and nonsmokers
gene: CREB5 | 205931_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2027
p=0.8887
p=0.2341
p=0.0362
p=0.9564
p=0.7884
p=0.7232
p=0.5702
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

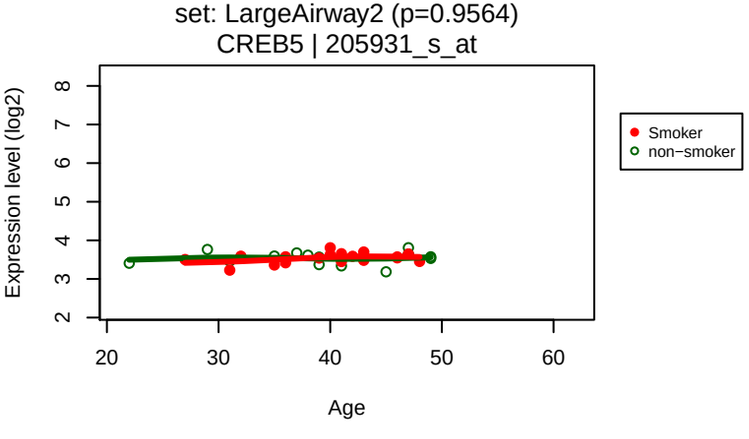

Gene expression ratio (smokers/nonsmokers)
gene: CREB5 | 205931_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2027
p=0.8887
p=0.2341
p=0.0362
p=0.9564
p=0.7884
p=0.7232
p=0.5702
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

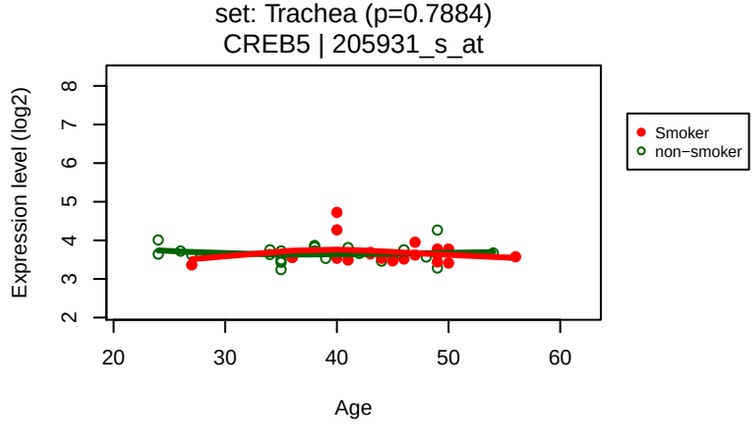

Gene expression ratio in Males
gene: CREB5 | 205931_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7942
p=0.1975
p=0.9612
p=0.6825
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB5 | 205931_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9091
p=0.2165
p=0.3808
p=0.6809
SmallAirway2
LargeAirway2
Trachea
Macrophage2

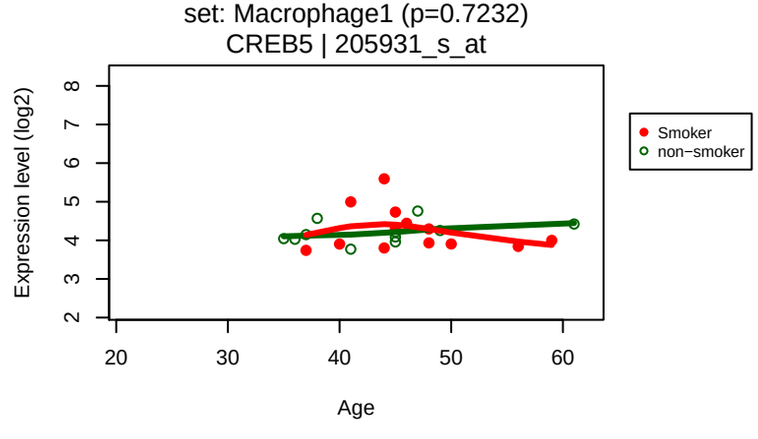

set: SmallAirway1 (p=0.2027)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker

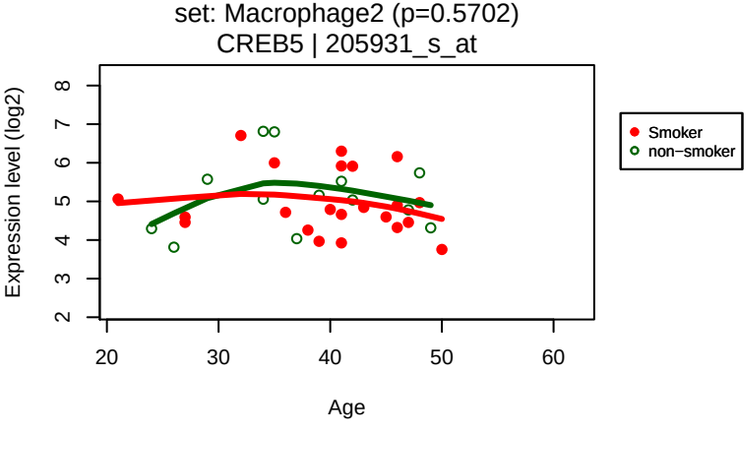

set: SmallAirway2 (p=0.8887)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker

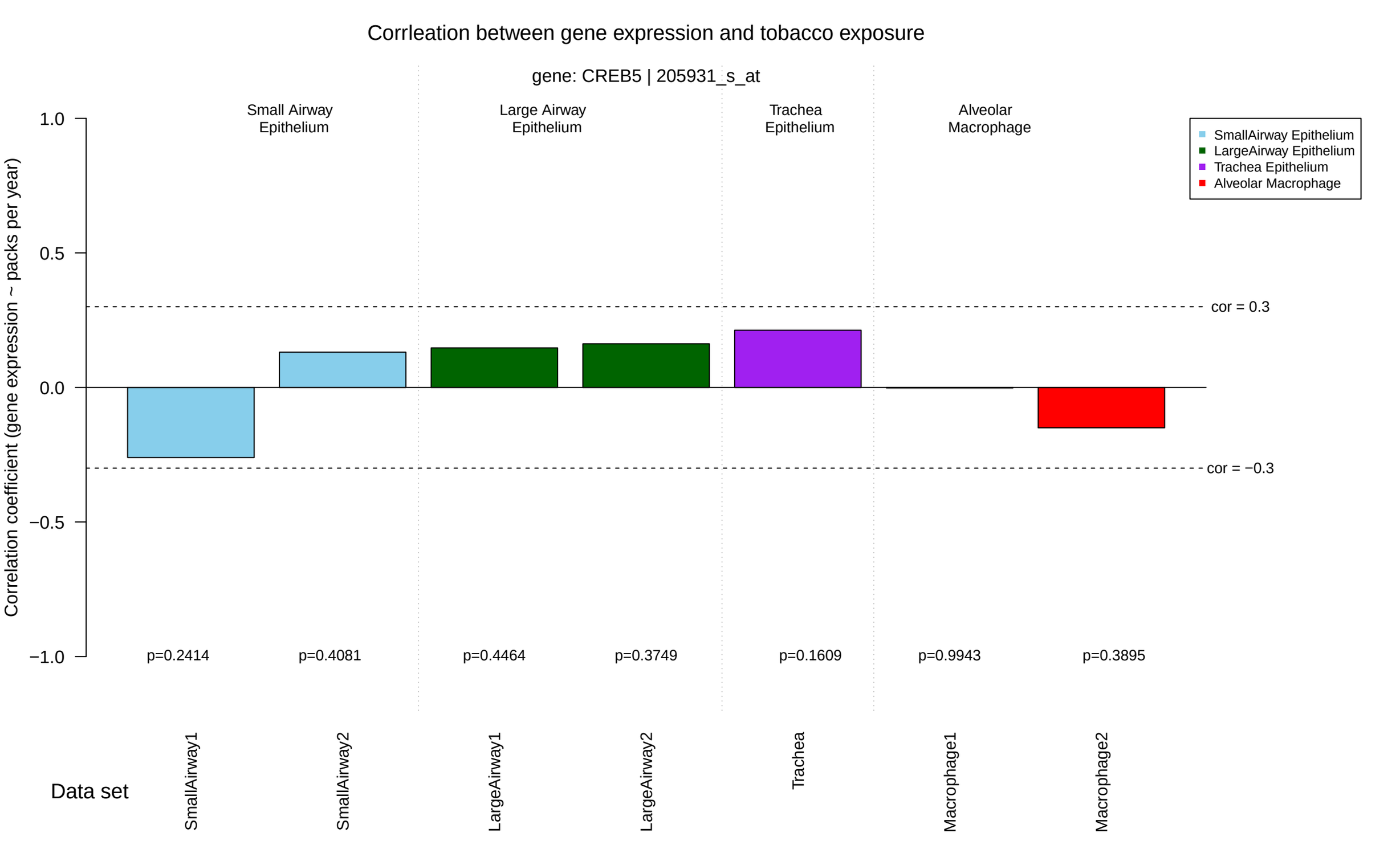

set: LargeAirway1 (p=0.0362)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9564)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7884)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7232)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5702)
CREB5 | 205931_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB5 | 205931_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2414
p=0.4081
p=0.4464
p=0.3749
p=0.1609
p=0.9943
p=0.3895
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2